

MIDTERM I—SOLUTIONS

ECO403/503—INTRODUCTION TO CONTRACT THEORY AND AUCTIONS

DECEMBER 2, 2020

ALEJANDRO MELO PONCE

NAZARBAYEV UNIVERSITY

Instructions. The exam has a total of 120 points. I will grade it over 100, so there is the possibility to leave some items unanswered and still get the top grade of 100; alternatively, there is also the chance of obtaining 20 extra points above the top grade. The points of each question are shown in boldface. *You need to submit your written solutions in pdf format to Moodle. The submission also consists of a short video, strictly less than 15 minutes, in which you record yourself explaining to me your solutions. The deadline is Monday, December 7 at 11:59:59 PM. Good luck!*

1. Consider the optimal regulatory scheme for a regulator of a monopolist who has known demand function $x(p)$, with $x'(p) < 0$ (i.e., demand is downward sloping), and a privately observed constant marginal cost of production θ . The regulator can set the monopolist's price and can make a transfer from or to the monopolist, so the set of regulatory schemes is $X = \{(p, t) : p > 0, t \in \mathbb{R}\}$. The regulator must guarantee the monopolist a nonnegative profit regardless of his production cost to prevent the monopolist from shutting down. The monopolist's marginal cost θ is drawn from $[\underline{\theta}, \bar{\theta}]$ with $\bar{\theta} > \underline{\theta} > 0$ according to the cumulative distribution function $\Phi(\cdot)$, which has an associated strictly positive probability density function $\phi(\cdot) > 0$. Assume that $\Phi(\theta)/\phi(\theta)$ is nondecreasing in θ . Denote a type- θ monopolist's profits from the scheme (p, t) by $\pi(p, t, \theta) = (p - \theta)x(p) + t$.

Suppose that the regulator wants to implement a direct revelation regulatory scheme $(p(\theta), t(\theta))$, i.e., a regulatory scheme which will induce the monopolist to reveal his true type. Define $\Pi(\theta)$ as

$$\Pi(\theta) = (p(\theta) - \theta)x(p(\theta)) + t(\theta).$$

- (a) (2 points) Write the participation constraints of the monopolist in terms of $\Pi(\theta)$.
(b) (3 points) Write the incentive compatibility constraints of the monopolist in terms of $\Pi(\theta)$.
(c) (15 points) Prove the following theorem:

Theorem. A regulatory scheme $(p(\theta), t(\theta))$ is incentive compatible if and only if

i. $p(\theta)$ is increasing.

ii. $\Pi(\theta) = \Pi(\theta') - \int_{\theta'}^{\theta} x(p(\tau))d\tau$ for any θ and θ' .

- (d) (3 points) Suppose that the regulator wants to design a direct revelation regulatory scheme that maximizes the expected value of a weighted sum of consumer and producer surplus minus the cost of transfers that the regulator makes to the monopolist, i.e., the regulator wants to maximize

$$\int_{\underline{\theta}}^{\bar{\theta}} \left(\int_{p(\theta)}^{\infty} x(s)ds + \alpha \pi(p(\theta), t(\theta), \theta) - t(\theta) \right) \phi(\theta)d\theta.$$

where $\alpha < 1$. State (but do not solve) the regulator's screening problem.

- (e) (15 points) Characterize as far as you can the regulator's optimal regulatory scheme.
(f) (2 points) What happens when $\alpha \geq 1$?

Facts. The Fundamental Theorem of Calculus. This might prove useful to remember. Let f be a continuous real-valued function defined on the closed interval $[a, b]$. Let F be the function defined, for all $x \in [a, b]$ by $F(x) = \int_a^x f(\tau)d\tau$. Then F is uniformly continuous on $[a, b]$ and differentiable on the open interval (a, b) , and $F'(x) = f(x)$ for all $x \in (a, b)$.

A corollary of the FTC. If we define the function G as $G(x) = \int_x^b f(\tau)d\tau$, then $G'(x) = -f(x)$ for all $x \in (a, b)$.

Solution

- (a) The participation constraints are $\Pi(\theta) \geq 0 \forall \theta \in [\underline{\theta}, \bar{\theta}]$.
(b) The incentive compatibility constraints are

$$\Pi(\theta) \geq \pi(p(\theta'), t(\theta'), \theta) \forall \theta', \theta \text{ in } [\underline{\theta}, \bar{\theta}],$$

or alternatively,

$$\Pi(\theta) = \max_{\theta' \in [\underline{\theta}, \bar{\theta}]} \{\pi(p(\theta'), t(\theta'), \theta)\} \forall \theta \in [\underline{\theta}, \bar{\theta}]$$

- (c) *Proof.* I begin with the “only if” ($\Leftarrow$) direction. Let $\theta' > \theta$, then incentive compatibility for these two types can be written as

$$\begin{aligned} \Pi(\theta) &\geq \Pi(\theta') + x(p(\theta))(\theta' - \theta) \\ \Pi(\theta') &\geq \Pi(\theta) - x(p(\theta))(\theta' - \theta) \end{aligned}$$

which implies that

$$(\theta' - \theta)x(p(\theta')) \leq \Pi(\theta) - \Pi(\theta') \leq (\theta' - \theta)x(p(\theta)).$$

The fact that demand is downward sloping, i.e. $x' < 0$ for any price, and that $\theta' > \theta$ implies that $p(\theta) < p(\theta')$ which in turn implies that p is increasing. Monotonicity of p implies that it is continuous almost everywhere. Dividing the previous expression by $(\theta' - \theta)$ yields

$$x(p(\theta')) \leq \frac{\Pi(\theta) - \Pi(\theta')}{\theta' - \theta} \leq x(p(\theta)).$$

Taking limits as $\theta' \rightarrow \theta$ yields $\frac{d\Pi(\theta)}{d\theta} = -x(p(\theta))$ at all points where p is continuous. Since Π is absolutely continuous then

$$\Pi(\theta) = \Pi(\theta') - \int_{\theta'}^{\theta} x(p(\tau))d\tau, \forall \theta, \theta'.$$

This concludes the only if part.

For the “if” ($\Rightarrow$) direction, consider any $\theta' > \theta$. Then

$$\begin{aligned} \Pi(\theta) - \Pi(\theta') &= \int_{\theta}^{\theta'} x(p(\tau))d\tau \\ &\geq \int_{\theta}^{\theta'} x(p(\theta'))d\tau \\ &= x(p(\theta'))(\theta' - \theta) \end{aligned}$$

where the inequality follows from p increasing and the downward sloping demand. □

- (d) The problem is not as “tricky” as it might seem to be on a first impression. Let me rewrite it a little bit so that it looks more standard. Let consumer surplus be defined, for any price p , as

$$CS(p) = \int_p^{\infty} x(s)ds.$$

Then the regulator’s problem is

$$\max_{p(\theta), t(\theta)} \int_{\underline{\theta}}^{\bar{\theta}} (CS(p(\theta)) + \alpha \pi(p(\theta), t(\theta), \theta) - t(\theta)) \phi(\theta) d(\theta) \quad (1)$$

$$\text{subject to } \Pi(\theta) \geq 0 \forall \theta \in [\underline{\theta}, \bar{\theta}] \quad (2)$$

$$\Pi(\theta) \geq \pi(p(\theta'), t(\theta'), \theta) \forall \theta', \theta \text{ in } [\underline{\theta}, \bar{\theta}] \quad (3)$$

(e) I start by simplifying the participation constraints. Notice that from the integral form of incentive compatibility and the participation constraint for the highest type, we have that

$$\Pi(\bar{\theta}) = \Pi(\underline{\theta}) - \int_{\underline{\theta}}^{\bar{\theta}} x(p(\tau))d\tau \geq 0. \quad (4)$$

The previous equation implies that

$$\Pi(\theta) = \Pi(\underline{\theta}) - \int_{\underline{\theta}}^{\theta} x(p(\tau))d\tau \geq \int_{\underline{\theta}}^{\bar{\theta}} x(p(\tau))d\tau - \int_{\underline{\theta}}^{\theta} x(p(\tau))d\tau = \int_{\theta}^{\bar{\theta}} x(p(\tau))d\tau \geq 0$$

Thus, the participation constraint of the highest type implies all the other participation constraints. Therefore, the problem is equivalent to

$$\max_{p(\theta), t(\theta)} \int_{\underline{\theta}}^{\bar{\theta}} (CS(p(\theta)) + \alpha\pi(p(\theta), t(\theta), \theta) - t(\theta))\phi(\theta)d\theta \quad (5)$$

$$\text{subject to } \Pi(\bar{\theta}) \geq 0 \quad (6)$$

$$\Pi(\theta) = \Pi(\underline{\theta}) - \int_{\underline{\theta}}^{\theta} x(p(\tau))d\tau \quad \forall \theta \quad (7)$$

$$p \text{ is increasing} \quad (8)$$

Notice that from the incentive compatibility constraints we have:

$$\begin{aligned} t(\theta) &= \Pi(\underline{\theta}) - \int_{\underline{\theta}}^{\theta} x(p(\tau))d\tau - (p(\theta) - \theta)x(p(\theta)) \\ &= \Pi(\bar{\theta}) + \int_{\underline{\theta}}^{\bar{\theta}} x(p(\tau))d\tau - \int_{\underline{\theta}}^{\theta} x(p(\tau))d\tau - (p(\theta) - \theta)x(p(\theta)) \\ &= \Pi(\bar{\theta}) + \int_{\theta}^{\bar{\theta}} x(p(\tau))d\tau - (p(\theta) - \theta)x(p(\theta)) \end{aligned} \quad (9)$$

Now let me focus on the term $\int_{\underline{\theta}}^{\bar{\theta}} (\alpha\pi(p(\theta), t(\theta), \theta) - t(\theta))\phi(\theta)d\theta$. First let me substitute the expression for the transfer (9) that we found out into the integrand:

$$\begin{aligned} \alpha\pi(p(\theta), t(\theta), \theta) - t(\theta) &= \alpha((p(\theta) - \theta)x(p(\theta)) + t(\theta)) - t(\theta) \\ &= \alpha(p(\theta) - \theta)x(p(\theta)) - (1 - \alpha)t(\theta) \\ &= \alpha(p(\theta) - \theta)x(p(\theta)) - (1 - \alpha)\left(\Pi(\bar{\theta}) + \int_{\theta}^{\bar{\theta}} x(p(\tau))d\tau - (p(\theta) - \theta)x(p(\theta))\right) \\ &= (p(\theta) - \theta)x(p(\theta)) - (1 - \alpha)\Pi(\bar{\theta}) - (1 - \alpha)\int_{\theta}^{\bar{\theta}} x(p(\tau))d\tau \end{aligned} \quad (10)$$

Then we have that:

$$\begin{aligned} &\int_{\underline{\theta}}^{\bar{\theta}} (\alpha\pi(p(\theta), t(\theta), \theta) - t(\theta))\phi(\theta)d\theta \\ &= \int_{\underline{\theta}}^{\bar{\theta}} (p(\theta) - \theta)x(p(\theta))\phi(\theta)d\theta - (1 - \alpha)\Pi(\bar{\theta}) - (1 - \alpha)\int_{\underline{\theta}}^{\bar{\theta}} \left(\int_{\theta}^{\bar{\theta}} x(p(\tau))d\tau\right)\phi(\theta)d\theta \end{aligned}$$

Using integration by parts ($u = \int_{\theta}^{\bar{\theta}} x(p(\tau))d\tau$) we can simplify the last term as:

$$\int_{\underline{\theta}}^{\bar{\theta}} \left(\int_{\theta}^{\bar{\theta}} x(p(\tau))d\tau\right)\phi(\theta)d\theta = \int_{\underline{\theta}}^{\bar{\theta}} x(p(\theta))\Phi(\theta)d\theta$$

Therefore, the objective (5) becomes:

$$\int_{\underline{\theta}}^{\bar{\theta}} \left\{ CS(p(\theta)) + (p(\theta) - \theta)x(p(\theta)) - (1 - \alpha)x(p(\theta)) \frac{\Phi(\theta)}{\phi(\theta)} \right\} \phi(\theta) d\theta - (1 - \alpha)\Pi(\bar{\theta}) \quad (11)$$

For $\alpha < 1$ the principal's problem reduces to

$$\begin{aligned} & \max_{q(\theta)} \int_{\underline{\theta}}^{\bar{\theta}} \left\{ CS(p(\theta)) + (p(\theta) - \theta)x(p(\theta)) - (1 - \alpha)x(p(\theta)) \frac{\Phi(\theta)}{\phi(\theta)} \right\} \phi(\theta) d\theta \\ & \text{subject to } p \text{ increasing} \end{aligned}$$

Ignoring momentarily the monotonicity constraint, we have that the first order condition with respect to p for each θ is

$$CS'(p(\theta)) + (p(\theta) - \theta)x'(p(\theta)) + x(p(\theta)) - (1 - \alpha)x'(p(\theta)) \frac{\Phi(\theta)}{\phi(\theta)} = 0$$

since $CS'(p(\theta)) = -x(p(\theta))$ we have that

$$\begin{aligned} -x(p(\theta)) + (p(\theta) - \theta)x'(p(\theta)) + x(p(\theta)) - (1 - \alpha)x'(p(\theta)) \frac{\Phi(\theta)}{\phi(\theta)} &= 0 \\ (p(\theta) - \theta)x'(p(\theta)) - (1 - \alpha)x'(p(\theta)) \frac{\Phi(\theta)}{\phi(\theta)} &= 0 \end{aligned}$$

dividing by $x'(p(\theta))$, which is not equal to zero, yields

$$p(\theta) = \theta + (1 - \alpha) \frac{\Phi(\theta)}{\phi(\theta)} \quad (12)$$

Since the problem assumes that $\frac{\Phi(\theta)}{\phi(\theta)}$ is nondecreasing in θ we have that indeed $p(\theta)$ is increasing in θ , which satisfies the monotonicity constraint. Transfers $t(\theta)$ then are given by (9):

$$t(\theta) = \int_{\underline{\theta}}^{\bar{\theta}} x(p(\tau)) d\tau - (p(\theta) - \theta)x(p(\theta)).$$

- (f) If $\alpha = 1$ then from (12) we get that $p(\theta) = \theta$, i.e., the price the regulator allows the monopolist to charge is equal to marginal cost. On the other hand, if $\alpha > 1$ then from the objective of the regulator (1) we have that the problem has *no solution*, since for any transfer function $t(\cdot)$ we can increase the objective (weighted sum of surplus) by raising it to $t(\cdot) + \epsilon$ for any ϵ .

2. A risk neutral principal hires a risk averse agent to undertake a project, which has a fixed cost of $I \geq 0$ for the principal. The project is either successful, in which case the value of the project for the principal is 20, or it is a failure, in which case the value is 4. The agent chooses to put in either high or low effort, i.e. $e \in \{0, 1\}$ where $e = 0$ corresponds to low effort, and $e = 1$ corresponds to high effort. If he puts in high effort, then the probability of success is 0.75, and if he chooses low effort, the probability of success is 0.25. The principal cannot observe the agent's effort but observes whether the outcome is a success or a failure. Therefore, feasible contracts are payments to the agent in case of success or failure, i.e., (w_s, w_f) . Therefore, the principal's payoff is $20 - w_s$, in case of success, and $4 - w_f$, in case of failure. Assume that the principal's payoff if she does not invest in the project is zero.

The agent has a utility function given by $\ln(w) - c(e)$, where $c(1) = \ln(4)$ and $c(0) = 0$. Also assume that the reservation utility of the agent (i.e., the payoff he gets if he does not accept the contract offer) is zero.

Facts (Some properties about $\ln$ and e). Here are some useful facts:

- $\lim_{x \rightarrow 0} \ln(x) = -\infty$
- $\ln(e) = 1, \ln(4) > 0, a \ln(x) = \ln(x^a), \ln(a) + \ln(b) = \ln(ab), e^{\ln(x)} = x$.
- $\frac{d \ln(x)}{dx} = \frac{1}{x}$

- (a) (12 points) What is the optimal contract that induces high effort?
- (b) (12 points) What is the optimal contract that induces low effort?
- (c) (4 points) What effort level is optimal to induce?
- (d) (12 points) Assume that it is optimal to induce high effort. For what range of I , the principal will find it optimal to hire the agent? How about if the principal could observe and contract on effort?

Solution

- (a) The principal's problem is

$$\begin{aligned} \max_{w_s, w_f} \quad & 0.75(20 - w_s) + 0.25(4 - w_f) - I \\ \text{subject to} \quad & 0.75 \ln(w_s) + 0.25 \ln(w_f) - \ln(4) \geq 0 \\ & 0.75 \ln(w_s) + 0.25 \ln(w_f) - \ln(4) \geq 0.25 \ln(w_s) + 0.75 \ln(w_f) \end{aligned}$$

Note that the participation constraint implies that $w_f, w_s > 0$, whereas the incentive compatibility constraint implies that $w_s > w_f$. Thus, we don't need to add non-negativity constraints. The Lagrangian is given by

$$\mathcal{L} = 0.75(20 - w_s) + 0.25(4 - w_f) - I + \mu[0.75 \ln(w_s) + 0.25 \ln(w_f) - \ln(4)] + \lambda[0.5(\ln(w_s) - \ln(w_f)) - \ln(4)]$$

The first order conditions with respect to w_s and w_f are

$$-0.75 + \frac{0.75\mu}{w_s} + \frac{0.5\lambda}{w_s} = 0 \quad (13)$$

$$-0.25 + \frac{0.25\mu}{w_f} - \frac{0.5\lambda}{w_f} = 0 \quad (14)$$

Condition (14) together with $w_f > 0$ and $\lambda \geq 0$ imply that $\mu > 0$. Therefore the participation constraint is binding. Adding up (13) and (14) yields $\mu = 0.75w_s + 0.25w_f$. substituting this (14) yields that $\lambda = \frac{3}{8}(w_s - w_f) > 0$ which implies that the IC constraint is also binding. Therefore, we can use the PC and IC constraints as equalities to solve for w_f and w_s as

$$w_f = \frac{1}{2}, \quad w_s = 8$$

(b) The principal's problem is

$$\begin{aligned} \max_{w_s, w_f} \quad & 0.25(20 - w_s) + 0.75(4 - w_f) - I \\ \text{subject to} \quad & 0.25 \ln(w_s) + 0.75 \ln(w_f) \geq 0 \\ & 0.25 \ln(w_s) + 0.75 \ln(w_f) \geq 0.75 \ln(w_s) + 0.25 \ln(w_f) - \ln(4) \end{aligned}$$

Note that the participation constraint implies that $w_f, w_s > 0$. Thus, we do not need to add non-negativity constraints.

The Lagrangian is

$$\mathcal{L} = 0.25(20 - w_s) + 0.75(4 - w_f) - I + \mu[0.25 \ln(w_s) + 0.75 \ln(w_f)] + \lambda[0.5(\ln(w_f) - \ln(w_s)) + \ln(4)]$$

The first order conditions with respect to w_s and w_f are

$$-0.25 + \frac{0.25\mu}{w_s} - \frac{0.5\lambda}{w_s} = 0 \quad (15)$$

$$-0.75 + \frac{0.75\mu}{w_f} + \frac{0.5\lambda}{w_f} = 0 \quad (16)$$

Condition (15), $\lambda \geq 0$ and $w_s > 0$ imply that $\mu > 0$. Therefore the participation constraint is binding. Again, adding up (15) and (16) yields $\mu = 0.25w_s + 0.75w_f$. Substituting this into (15) yields that $\lambda = \frac{3}{8}(w_f - w_s)$. Since $\lambda \geq 0$, the previous result implies that $w_f \geq w_s$.

We will consider two cases. Suppose that $w_f > w_s$ so that $\lambda > 0$. Then the IC constraint also binds. Using the binding PC and IC constraints as equalities to solve for w_f and w_s yields once again that $w_s = 8$ and $w_f = 1/2$; however, this is incompatible with $\lambda > 0$ so we get a contradiction. Thus, we must have $w_f = w_s$ which implies that $\lambda = 0$.

The fact that $\lambda = 0$ and $w_f = w_s$, together with the binding PC constraint implies that $w_s = w_f = 1$.

(c) Expected profit under high effort is $9.875 - I$, whereas under low effort is $7 - I$. Clearly, it is optimal to induce high effort.

(d) The principal will hire an agent if $9.875 - I \geq 0$ or $I \leq 9.875$.

If the principal could observe effort, then the only constraint that matters is the PC constraint. Therefore, if the principal wants to induce high effort, her problem is

$$\begin{aligned} \max_{w_s, w_f} \quad & 0.75(20 - w_s) + 0.25(4 - w_f) - I \\ \text{subject to} \quad & 0.75 \ln(w_s) + 0.25 \ln(w_f) - \ln(4) \geq 0 \end{aligned}$$

Once again, notice that the PC constraint implies that w_s and w_f are strictly positive. The Lagrangian is given by

$$\mathcal{L} = 0.75(20 - w_s) + 0.25(4 - w_f) - I + \mu[0.75 \ln(w_s) + 0.25 \ln(w_f) - \ln(4)]$$

The first order conditions with respect to w_s and w_f are

$$-0.75 + \frac{0.75\mu}{w_s} = 0 \quad (17)$$

$$-0.25 + \frac{0.25\mu}{w_f} = 0 \quad (18)$$

From this we conclude that $\mu = w_f = w_s > 0$ and the PC constraint binds. Therefore

$$\ln(w^*) = \ln(4) \Rightarrow w^* = w_s = w_f = 4.$$

Expected profit is equal to $12 - I$. As before, expected profit under low effort is $7 - I$ (because in the second best only the PC constraint binds, so the solution to the first-best low effort is the same). Therefore, when effort is observable, inducing high effort is optimal (no need to show this since I asked you just to assume it) and the principal finds it optimal to hire an agent if $I \leq 12$. This implies that projects that cost $I \in (9.875, 12)$ are not undertaken due to moral hazard.

3. A risk neutral landowner is going to offer a tenancy contract to a financially constrained tenant. The crop yield is either high, q_h , or low q_l , with $q_h > q_l \geq 0$. The tenant chooses to put in either high or low effort. If he puts in high effort, then the probability of high output is p_h , and if he chooses low effort, the probability of high output is p_l , with $p_h > p_l > 0$. The landlord cannot observe the tenant's effort but observes whether the output is low or high, and hence feasible contracts are transfers to the tenant in case of high and low outputs, i.e., (t_h, t_l) . Therefore, the landlord's payoff is $q_h - t_h$, if output is high, and $q_l - t_l$, if output is low. If the tenant receives a transfer t and puts in high effort, then his payoff is $t - c$, where $c > 0$ is the cost of high effort. If he chooses low effort, then his payoff is simply t . The tenant has limited liability so that $t_l \geq 0$ and $t_h \geq 0$. Also assume that the reservation utility (i.e., the payoff he gets if he does not accept the contract offer) of the tenant is zero.

- (9 points) What is the optimal contract that induces high effort?
- (9 points) What is the optimal contract that induces low effort?
- (4 points) What effort level is optimal to induce?
- (9 points) Assume that it is optimal to induce high effort. Suppose that at cost $p > 0$, the landlord can monitor and learn with certainty whether the agent puts in high or low effort. For what range of p , the landlord will find it optimal to monitor?
- (9 points) Now suppose that there is a functioning credit market so that there are no limited liability constraints anymore. Find the landlord's expected profit under the optimal contract and compare it with her expected profit from part (a).

Solution

- (a) The landlord's problem is

$$\begin{aligned}
 \max_{t_h, t_l} \quad & p_h(q_h - t_h) + (1 - p_h)(q_l - t_l) \\
 \text{subject to} \quad & p_h t_h + (1 - p_h)t_l - c \geq 0 & (\text{PC}) \\
 & p_h t_h + (1 - p_h)t_l - c \geq p_l t_h + (1 - p_l)t_l & (\text{IC}) \\
 & t_l \geq 0 & (LL_l) \\
 & t_h \geq 0 & (LL_h)
 \end{aligned}$$

Let me rewrite the IC constraint as

$$(p_h - p_l)(t_h - t_l) - c \geq 0$$

Firstly, the IC constraint implies that $t_h > t_l$. Therefore, LL_l implies LL_h , and hence, the latter limited liability constraint can be ignored. Secondly, either the PC constraint or the LL_l constraint must bind. Otherwise, the landlord can decrease t_l and t_h by the same amount without violating any of the constraints and increase profits.

We will now consider two cases. Suppose first that PC binds at the solution, i.e.,

$$t_h = \frac{c - (1 - p_h)t_l}{p_h}$$

Therefore,

$$\begin{aligned}
(p_h - p_l)(t_h - t_l) - c &= (p_h - p_l) \left(\frac{c - (1 - p_h)t_l}{p_h} - t_l \right) - c \\
&= \frac{p_h - p_l}{p_h} (c - t_l) - c \\
&= -\frac{p_h - p_l}{p_h} t_l - \frac{p_l}{p_h} c < 0
\end{aligned}$$

Thus, the IC constraint is violated so we have reached a contradiction. Therefore, the LL_l must bind, i.e., $t_l = 0$. This in turn, implies that the IC constraint must bind. Otherwise, since the PC constraint does not bind either, the landlord could increase profits by decreasing t_h by some small amount without violating any of the constraints.

Therefore, the optimal contract is

$$t_l = 0, \quad t_h = \frac{c}{p_h - p_l}$$

(b) The landlord's problem is

$$\begin{aligned}
&\max_{t_h, t_l} p_l(q_h - t_h) + (1 - p_l)(q_l - t_l) \\
&\text{subject to} \quad p_l t_h + (1 - p_l)t_l \geq 0 \quad (\text{PC}) \\
&\quad p_l t_h + (1 - p_l)t_l \geq p_h t_h + (1 - p_h)t_l - c \quad (\text{IC}) \\
&\quad t_l \geq 0 \quad (LL_l) \\
&\quad t_h \geq 0 \quad (LL_h)
\end{aligned}$$

Let me rewrite the IC constraint as

$$(p_h - p_l)(t_l - t_h) + c \geq 0$$

First suppose that $t_h > 0$. A strictly positive t_h implies that the PC constraint holds with strict inequality, i.e., is not binding. But if $t_h > 0$ this implies that the principal can lower it a little bit without violating any constraints and increase its profits. Therefore, we must have that $t_h = 0$. With the previous result, we can similarly conclude that if $t_l = 0$, then the principal can reduce it by some small amount without violating any of the constraints and increase her profits. Thus, $t_l = 0$.

We conclude that the optimal contract in this case is $t_l = t_h = 0$. Note that the limited liability constraints do not impose any burden with respect to the problem of the landlord without them, the solution would still be the same.

(c) The expected profit under high effort is

$$p_h \left(q_h - \frac{c}{p_h - p_l} \right) + (1 - p_h)q_l$$

and under low effort is

$$p_l q_h + (1 - p_l)q_l$$

The expected profit under high effort is higher if and only if

$$(p_h - p_l)(q_h - q_l) \geq \frac{c}{p_h - p_l}$$

If the previous condition holds, it is optimal to induce high effort, otherwise it is optimal to induce low effort.

(d) If the landlord is able to monitor by paying for the monitoring technology, then the only constraint that matters is the PC constraint, since the landlord can punish the tenant if he puts in an effort level

that she does not desire to induce. Therefore, if she wants to induce high effort, the landlord's problem is

$$\begin{aligned} \max_{t_h, t_l} \quad & p_h(q_h - t_h) + (1 - p_h)(q_l - t_l) - p \\ \text{subject to} \quad & p_h t_h + (1 - p_h)t_l - c \geq 0 \quad (\text{PC}) \\ & t_l \geq 0 \quad (LL_l) \\ & t_h \geq 0 \quad (LL_h) \end{aligned}$$

At the solution, the PC constraint must bind. If it doesn't, then $t_h > 0$ or $t_l > 0$, and hence the landlord can decrease t_l or t_h without violating any of the constraints and increase profit. Therefore,

$$p_h t_h + (1 - p_h)t_l - c = 0$$

together with $t_l \geq 0$ and $t_h \geq 0$ characterize the *solution set* to the landlord's problem. The expression in the display is the expected cost of any solution (t_l, t_h) that satisfies the previous conditions. Therefore, we can substitute this expected cost into the objective function to get the expected profit of the landlord:

$$p_h q_h + (1 - p_h)q_l - c - p$$

Therefore, monitoring is optimal if and only if

$$p_h q_h + (1 - p_h)q_l - c - p \geq p_h \left(q_h - \frac{c}{p_h - p_l} \right) + (1 - p_h)q_l$$

The previous expression is equivalent to

$$p \leq \frac{c p_l}{p_h - p_l}$$

(e) The landlord's problem is now

$$\begin{aligned} \max_{t_h, t_l} \quad & p_h(q_h - t_h) + (1 - p_h)(q_l - t_l) \\ \text{subject to} \quad & p_h t_h + (1 - p_h)t_l - c \geq 0 \quad (\text{PC}) \\ & p_h t_h + (1 - p_h)t_l - c \geq p_l t_h + (1 - p_l)t_l \quad (\text{IC}) \end{aligned}$$

At the solution, the PC constraint must bind, otherwise, the landlord could decrease t_l and t_h by the same small amount and increase her profit. Therefore,

$$p_h t_h + (1 - p_h)t_l - c = 0$$

Substituting this into the expected profit function we get

$$p_h q_h + (1 - p_h)q_l - c$$

The expected profit of the landlord when the tenant was financially constrained was

$$(p_h - p_l)(q_h - q_l) \geq \frac{c}{p_h - p_l}$$

Since $\frac{p_h}{p_h - p_l} > 1$ the expected profit when the tenant is not financially constrained is higher.